

SWS 3026 Visual Computing

Project Evaluation Criteria

Notes:

- A. *Your group project will be evaluated in 2 stages. In Stage 1, each group will take turns to give a 10-minute oral presentation using slides, followed by 3 minutes of Q&A.*
- B. *During the oral presentation, **do not run your demos/programs.** Instead, show screenshots or pre-recorded videos of your demos in action. **This rule will be strictly enforced.***
- C. *After all groups have presented, Stage 2 will begin. Here, all groups will give a demo of your program for the Bonus Level: the Dance Analysis. You will then view other groups' demos and vote for the best one. Details of the voting procedure will be given later.*
- D. *Only Stage 1 will be evaluated by the Instructors according to the criteria below. Each Instructor will score your work independently, and the combined score will count towards your final grade for the course. Stage 2 is a popularity contest and will not affect your final grade.*

Beginner level:

- 1. In Task 2, you used the Haar cascade detector. Show where this detector failed to work, ie. show screenshots or video of the failure cases.
- 2. Explain what you did in Task 3 to improve the detector. Did you perform image processing, or use a different detector model, or something else?
- 3. Show screenshots or video of your robust detector created in Task 3.

Expert level:

- 4. For Task 1, explain your pipeline to classify the 6 classes of facial expressions: what you did for pre-processing, for feature extraction, and your choice of the classifier model.
- 5. Show the confusion matrix and explain what problems it revealed, eg. the classes where your classifier performed poorly. Show some test images where your classifier misclassified or failed.
- 6. Explain how you overcame this problem: was it through pre-processing, or new features or more powerful classifiers? Show the confusion matrix after your improvement.
- 7. For Task 2, show screenshots or a video of your program.

8. Are you detecting additional facial expressions? What are these and how did you train your classifier to detect them?
9. Choose any 2 video effects that you implemented and explain how you went from detecting the facial expression to rendering the video effect.
10. Does your video effect exhibit any flickering, jerky animation, or other poor visual result? Show screenshots or a video of this problem. What is causing this, and how can/did you rectify the problem?

Bonus level:

11. For Task 1, show screenshots or a video of your program in action.
12. What challenges did you encounter in the detection and rendering of keypoints? How did you overcome these challenges?
13. For Task 2, show screenshots or a video of your Dance program in action.
14. How did you score the user's dance moves? How did you align the user's movements with the reference video? What similarity or distance metrics did you use?
15. Is your score numeric (eg. 9.2) or textual (eg. "Super") or both? How does the score change over time as the user is dancing? Do you compute an overall score after the dance is finished? How?
16. Besides dancing, what other actions can you detect and score? Explain how you render new objects (eg. flying fruits) if any. Also explain how you detect the new action (what features/training set you used), and how the score is computed. Use appropriate screenshots or videos to illustrate.